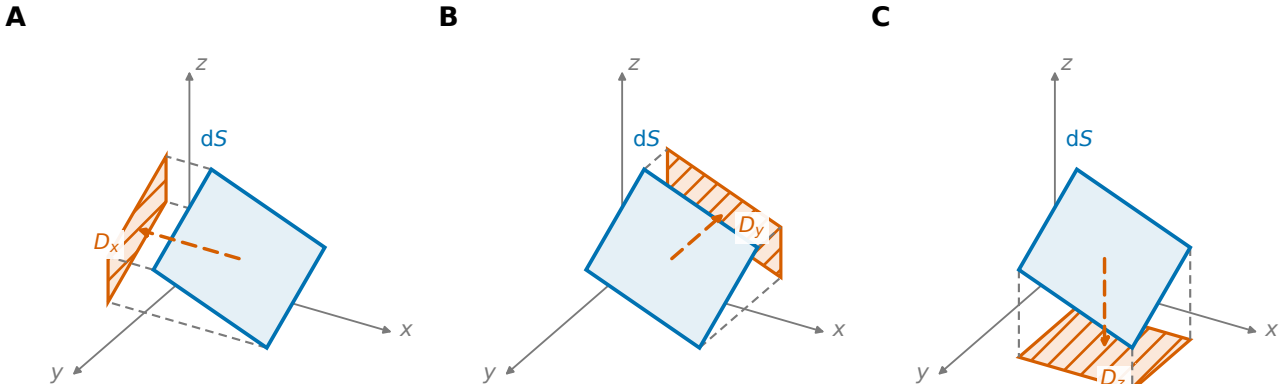

$dy\,dz = n_x\,\mathrm{d}S$

$dz\,dx = n_y\,\mathrm{d}S$

$dx\,dy = n_z\,\mathrm{d}S$

$$\mathrm{d}\mathbf{S} = (\mathrm{d}y\,\mathrm{d}z,\ \mathrm{d}z\,\mathrm{d}x,\ \mathrm{d}x\,\mathrm{d}y) = (n_x, n_y, n_z)\,\mathrm{d}S \qquad (n_x, n_y, n_z > 0)$$